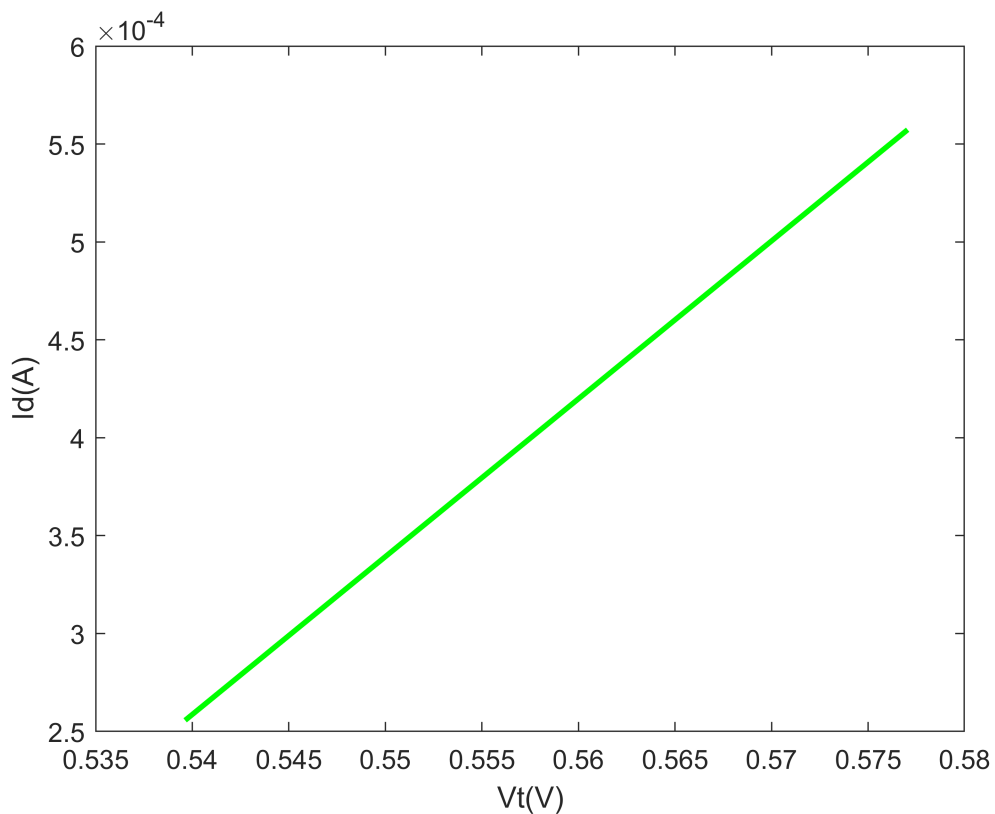


参数设置

```
clear;clf;
Is = 22.5e-6;n = 2;Vj = 0.4;L = 0.1;
filename = "PN 数据.csv";
table = readtable(filename);
V = table2array(table(:,1))';
I = table2array(table(:,2))';
```

通过上次实验数据求出 R_s (取 I_d 从 0.25mA 到 0.57mA)

```
p = polyfit(V(55:59),I(55:59),1);
k = p(1,1);%斜率
y=polyval(p,V(55:59));
plot(V(55:59),y,'g-','linewidth',2);
xlabel("Vt(V)");
ylabel("Id(A)");
```



$$R_s = 1/k$$

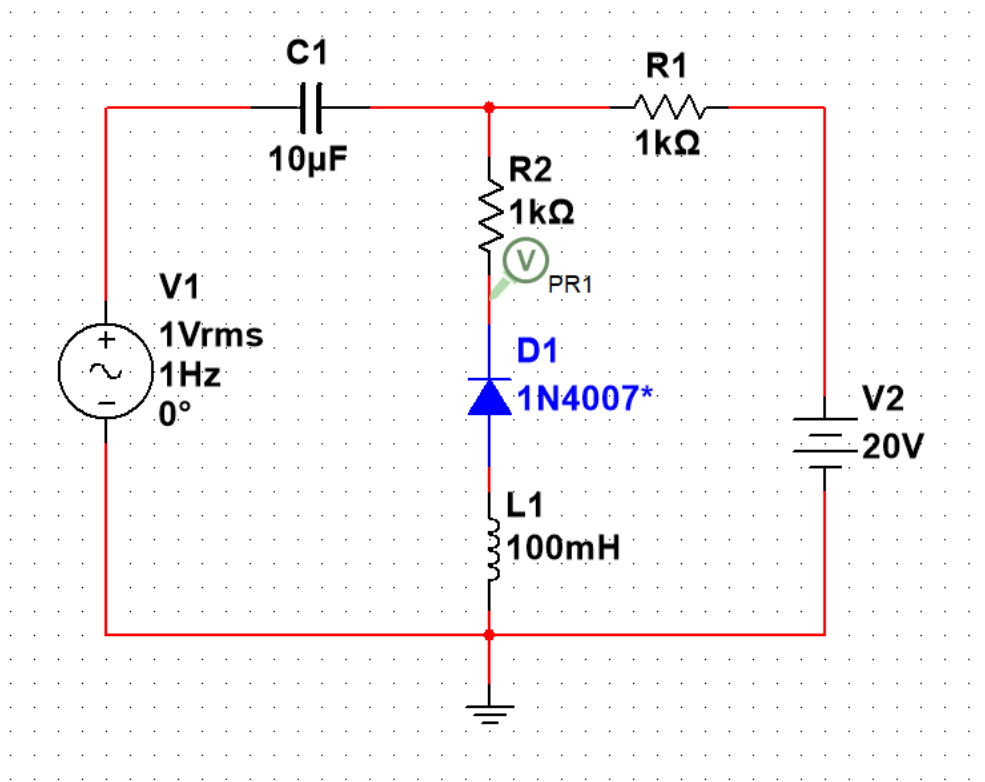
$$R_s = 124.0409$$

根据步骤 1, 假设 $C_j = 1\text{pF}$ 时, 求出谐振频率

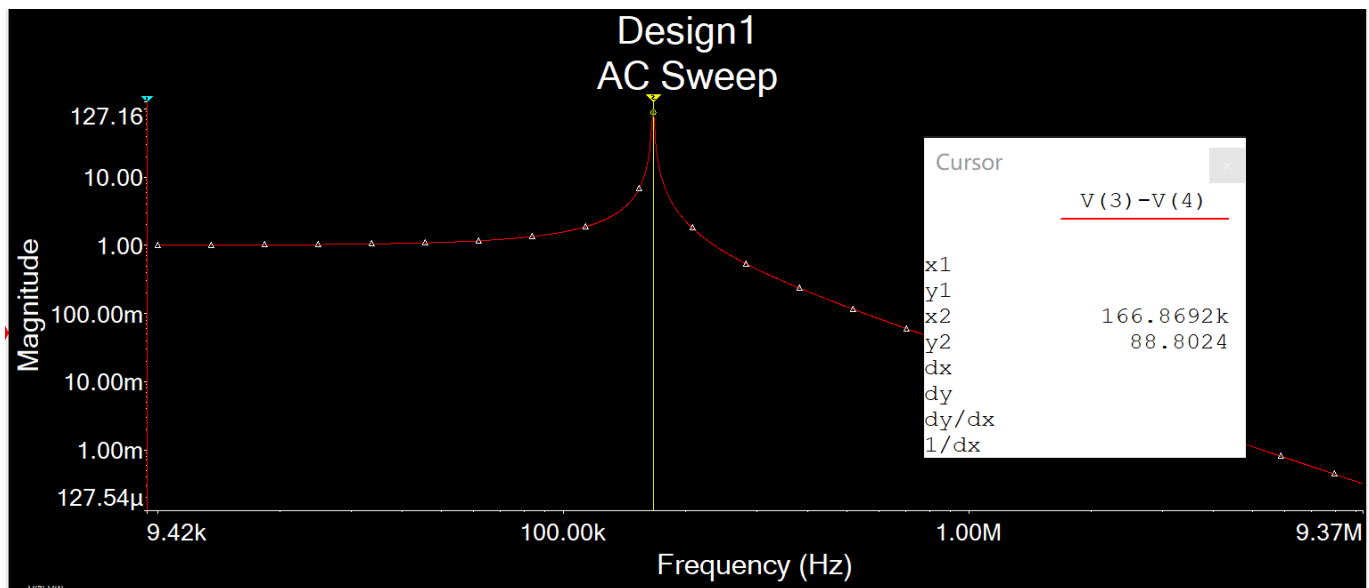
```
Cj = 1e-12;
w2 = 1/(L*Cj);
f0 = sqrt(w2)/(2*pi)
```

$f_0 = 5.0329 \times 10^5$

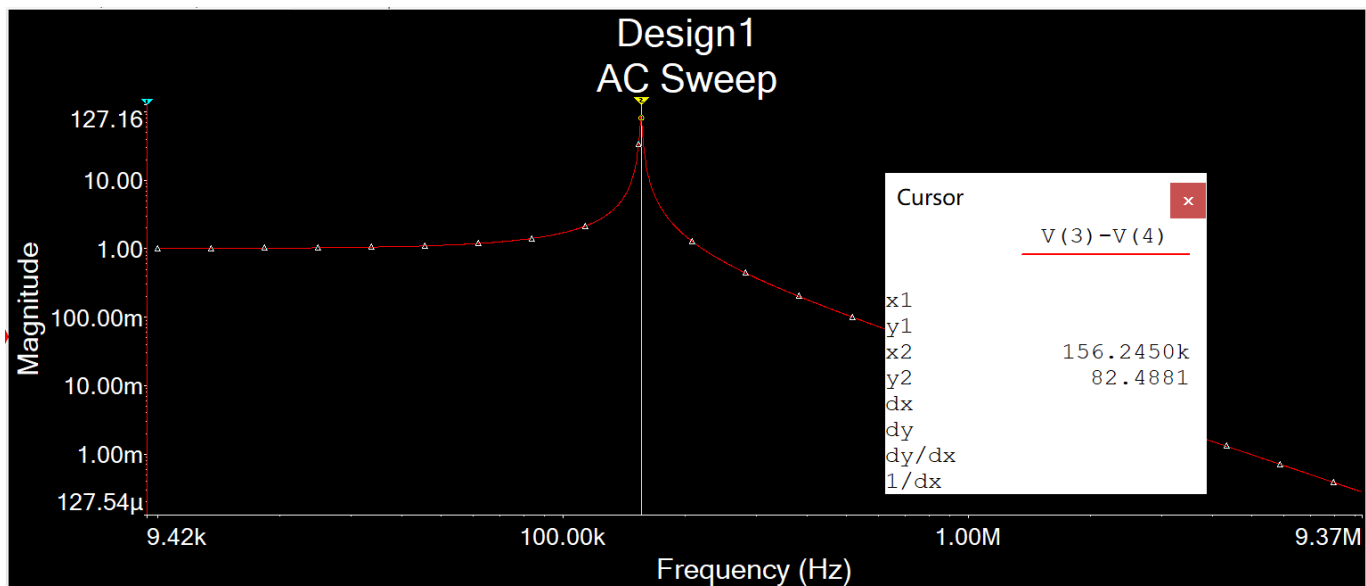
根据步骤二，搭建电路



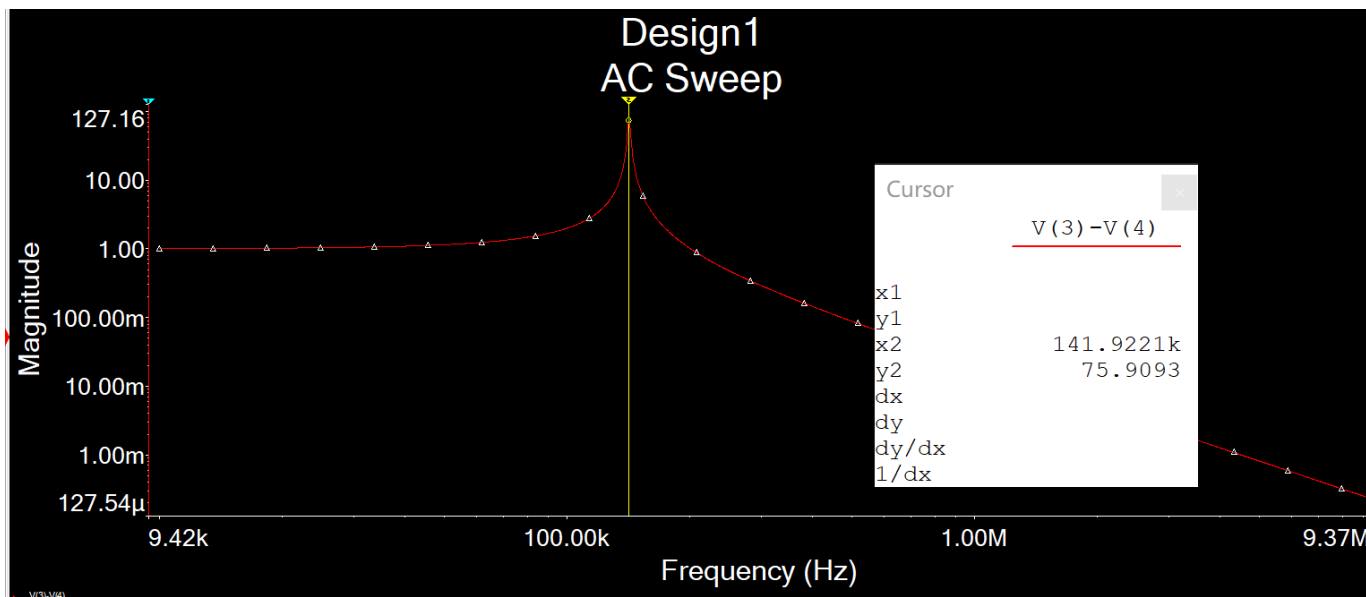
$V_d = 20V$



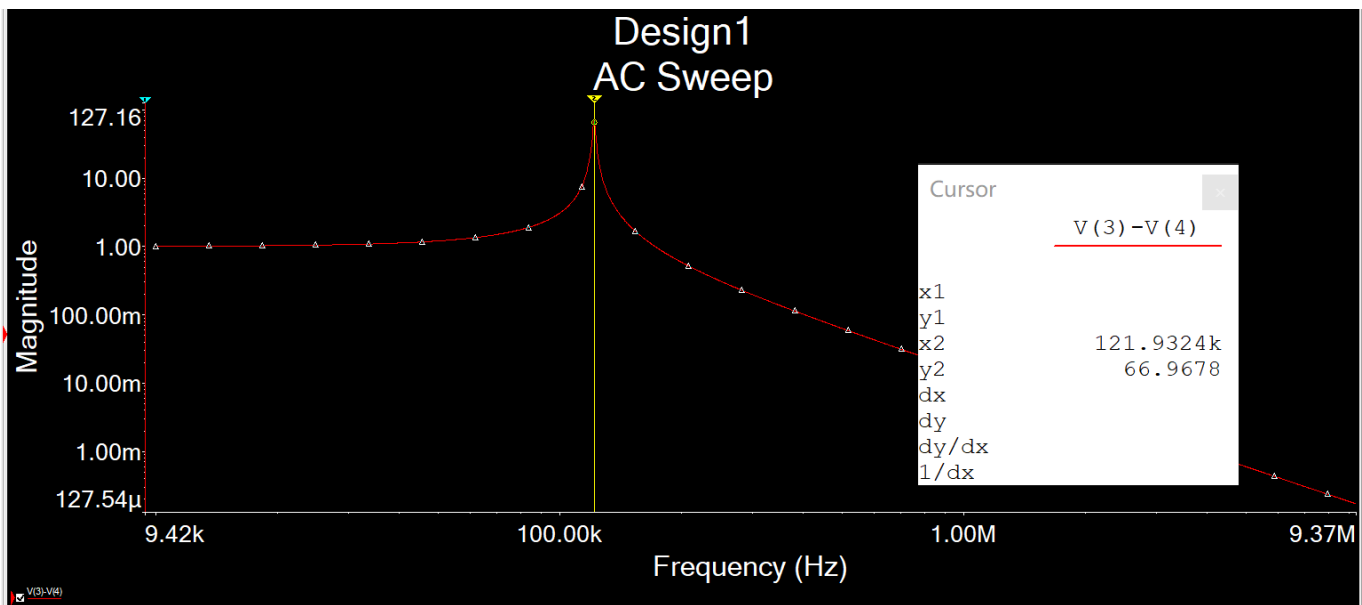
$V_d = 15V$



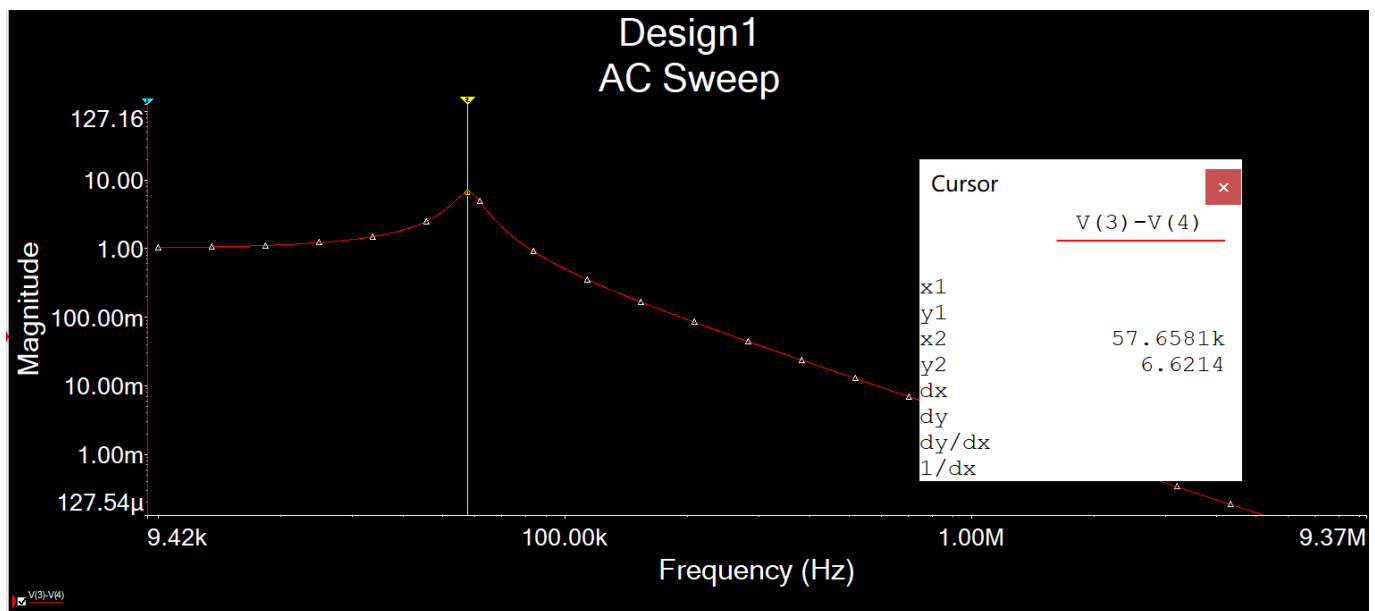
Vd = 10V



Vd = 5V



Vd = 0.3V



将每一个 Vd 对应的谐振频率 f 写入表格

根据公式 $C_j = \frac{1}{(2\pi f)^2 L}$ 计算 Cj 和 1/Cj^2

```
V = xlsread('Lab2 数据.xlsx', 'Sheet1', 'A2:A6');
f = xlsread('Lab2 数据.xlsx', 'Sheet1', 'B2:B6');
Cj = zeros(5,1);
Cj1 = zeros(5,1); % 1/(Cj^2)
for i = (1:5)
    Cj(i) = 1/((2*pi*f(i)*1000)^2*L);
    Cj1(i) = 1/(Cj(i)^2);
end
Cj
```

```
Cj = 5×1
10-10 x
    0.0910
    0.1038
    0.1258
    0.1704
    0.7619
```

Cj1

```
Cj1 = 5×1
1022 x
    1.2085
    0.9287
    0.6323
    0.3445
    0.0172
```

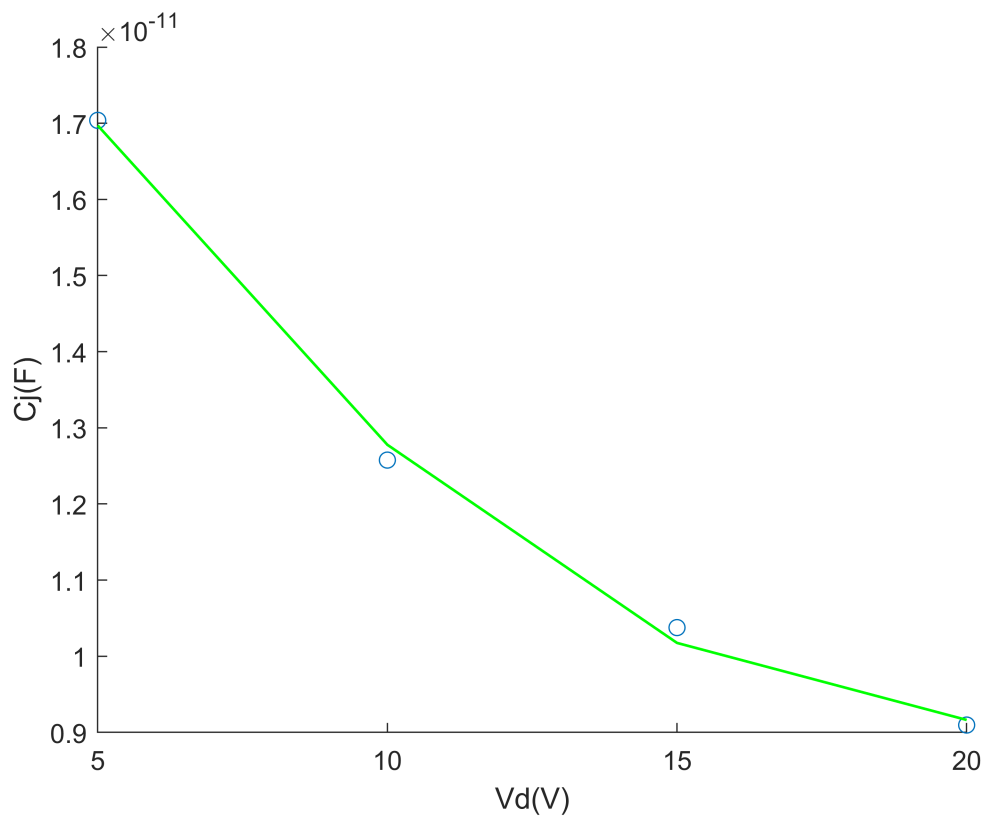
```
xlswrite('Lab2 数据.xlsx',Cj,'C2:C6');
xlswrite('Lab2 数据.xlsx',Cj1,'D2:D6');
```

由于在反向电压很小时，二极管不稳定，测量结果很不准确

舍去 $V_d = 0.3V$ 时的值

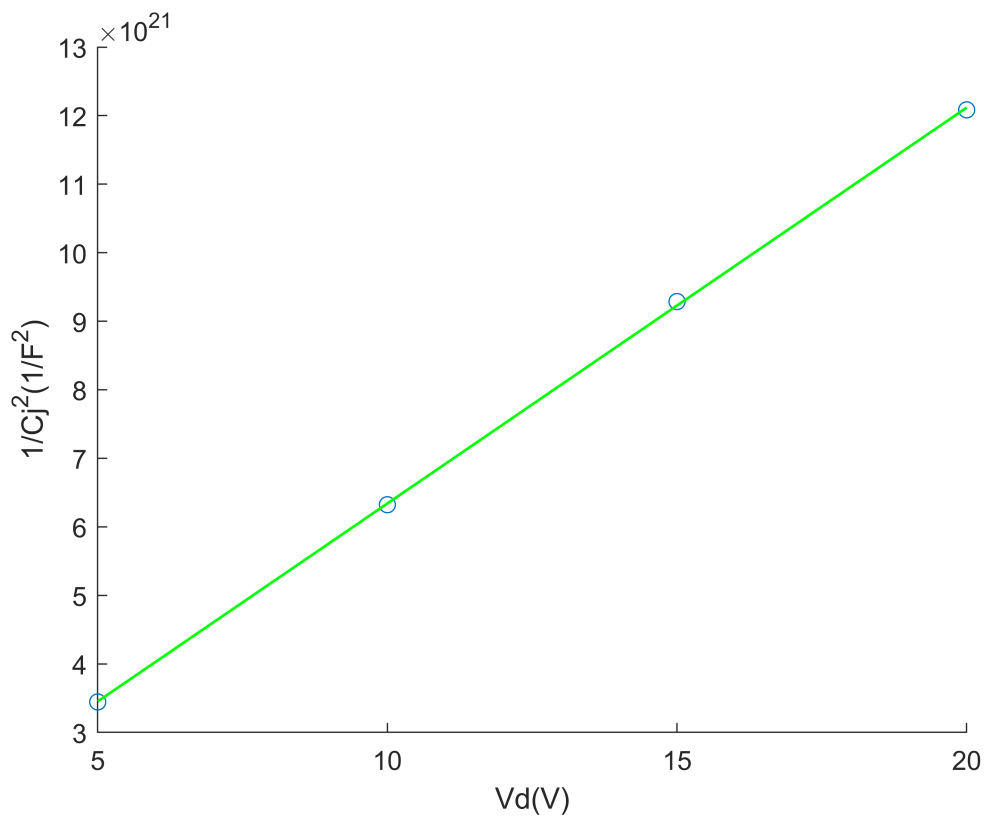
Cj — V_d 散点图+拟合曲线图

```
figure;
scatter(V(1:end-1),Cj(1:end-1));hold on;
p = polyfit(V(1:end-1),Cj(1:end-1),2);
y = polyval(p,V(1:end-1));
plot(V(1:end-1),y,'g-','linewidth',1);
xlabel("Vd(V)");
ylabel("Cj(F)");
```



1/Cj² — Vd 散点图+拟合直线图

```
figure;
scatter(V(1:end-1),Cj1(1:end-1));hold on;
p = polyfit(V(1:end-1),Cj1(1:end-1),1);
y = polyval(p,V(1:end-1));
plot(V(1:end-1),y,'g-','linewidth',1);
xlabel("Vd(V)");
ylabel("1/Cj^2(1/F^2)");
```



根据公式 $\frac{1}{C_j^2} = K(V_0 - V)$

当 $V = 0$ 时, $\frac{1}{C_{j0}^2} = KV_0$

得到 $C_{j0} = \sqrt{KV_0}$

```
k = p(1,1);%斜率
d = p(1,2);%截距
Cj0 = sqrt(1/d)
```

```
Cj0 = 4.2116e-11
```